MILITARY SPECIFICATION

PELLETS (PBX)
FOR USE IN AMMUNITION

1. SCOPE

1.1 This specification covers explosive pellets which are formed using a powdered explosive compound known as PBX (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-I-559 - Ink, Marking Stencil, Opaque, For Porous Surfaces (Wood Boxes, Fiber Cartons, Etc.)

MILITARY

MIL-P-14999- Powder, Molding Compound Explosive (PBX)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1376

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DRAWINGS

U. S. ARMY MUNITIONS COMMAND

8796522 - Marking Diagram and Sealing for Wood Packing Boxes
9211789 - Ink, Marking, Opaque for Porous Surfaces (For Automatic or Semi-Automatic Machine Appl.)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other Publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 792-66 - Specific Gravity and Density of Plastics by Displacement

(Requests for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 0-190

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402. Orders for the above publications should cite "49 CFR 0-190 (latest revision)").

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3. REQUIREMENTS

3.1 Material.-The PBX pellets shall conform to all the requirements specified on applicable drawings and referenced specifications.

3.2 Density.-The density of the PBX pellets shall conform to the requirements specified on applicable drawings when tested as specified in 4.4.1.

3.3 Composition.-The PBX pellets shall be homogeneous. The chemical composition shall be determined as specified in 4.4.2.

3.4 Workmanship.-The pellets shall be free of dirt, grease, and other foreign matter. The pellets shall be firmly compacted, free of soft spots and powdery areas.

3.5 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all requirements and quality assurance provisions specified in this specification.

b. Specification number and date, together with an identification and date of changes.

c. Certificates of analysis on all materials used directly by the contractor when such material is controlled by Government specifications shall be made available upon request by the Contracting Officer.

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d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principle. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample consisting of 100 PBX pellets in accordance with instructions issued by the Contracting Officer for evaluation in accordance with paragraph 4.2.2. All samples submitted shall have been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2 Inspections to be performed.-The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.3 and 4.4 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

4.3 *Inspection provisions*

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4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Each lot of PBX pellets shall be produced from one lot of PBX powder produced in accordance with MIL-P-14999 (MU). The product shall be submitted for inspection in accordance with MIL-STD-105. The criteria and procedure for the assignment of lot numbers shall be in accordance with Standard MIL-STD-1168.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105). Contractor's sampling plans, if used, shall be approved by the Government and shall provide as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed, using an AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each minor defect.

4.3.2.1 PBX pellets (see applicable drawing)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Weight	Balance	01001
102.	Length	Gage (CD)	01002
103.	Diameter	Gage (CD)	01003
Minor:	AQL 1.00 percent		
201.	Presence of cracks or laminations	Visual	01004
202.	Evidence of poor workmanship (see 3.4)	Visual	01005

4.3.3 Sampling.-A minimum of ten (10) pellets shall be removed from the beginning of the pelletizing operation and a minimum of 10 pellets shall be removed from the end of the pelletizing operation. The two 10 pellet samples shall be crushed and retained as two separate individual samples. These two samples shall be tested for

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the composition requirement in accordance with the test depicted in 4.4.2. If either of these samples fail to comply with the requirement specified in 3.3 the lot of PBX pellets shall be rejected. If the total weight of either of the samples described above (beginning or end) is less than 100 grams, then additional pellets shall be selected from the pelletizing operation (beginning and end) to obtain a minimum weight of 100 grams for each sample.

4.3.3.1 Density.-A random sample of ten (10) individual pellets shall be selected from the sample used in 4.3.2.1. If the density of these pellets fails to meet the requirement specified in 3.2, the lot shall be rejected.

4.4 Test Methods and Procedures

4.4.1 Density, Major Defect, Defect Code No. 02001.-The density shall be determined in accordance with ASTM Method D 792-66, Method A-1. Calculate the density of the PBX pellets as follows (see 6.5):

Density, minimum $\leq \overline{X} - 3s$

Where:

$\overline{X}$ = average of ten (10) individual pellets
s = standard deviation

4.4.2 Composition

RDX, Major Defect, Code No. 03001 - Dioctylphthalate, Major Defect, Code NO. 03002 - Polystyrene, Major Defect, Code No. 03003

The composition of the PBX molding powder shall be determined in accordance with the applicable paragraphs of MIL-P-14999 (MU).

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5. PREPARATION FOR DELIVERY

5.1 Packaging and Packing

5.1.1 Level C.-PBX pellets shall be packaged and packed in conformance with Code of Federal Regulations, 49 CFR, parts 0-190 and the manufacturer's standard practice. Pellets shall be packed in like quantities and containers shall be uniform.

5.2 Marking.-Each interior package shall be labeled, stamped or printed with the following information:

Dwg. (No. of applicable dwg.)
(QTY) (NOMENCLATURE)
LOT (Mfr. Symbol - No.)
Packed (Mo/Yr)

HIGH EXPLOSIVE - DANGEROUS

Letters shall be upper case and all marking minimum 1/4 inch in size. Labels shall be gummed white stock. Ink shall be black, conforming to TT-I-559 or drawing 9211789. Each exterior container shall be marked in accordance with drawing 8796522. DOT marking shall be HIGH EXPLOSIVES - DANGEROUS and shall appear on the top and one side of the box in letters not less than 7/16 inch high. The words THIS SIDE UP shall appear on the top of the exterior container. Letter size 7/16 inch minimum.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification
b. Description sheets shall be prepared for each lot in accordance with Standard MIL-STD-1171.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs and inspection data submitted by the contractor.

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6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers (on the Equipment Lists referenced on the Equipment Tabulation) or (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423).

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The contractor is referred to MIL-HDBK-204, "Inspection Equipment DXsign" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-A-P in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Intended use.-The PBX pellets covered in this specification are intended for use as booster charges to initiate high explosive.

6.5 "Engineering Design Handbook", AMC Pamphlet 706-110 has been found suitable for determining the average and standard deviation of the PBX pellets.

6.6 Military specification MIL-P-50977 (MU), dated 2 October 1972, supersedes purchase description PA-PD-2861, dated 5 May 1966 and shall be used for all future procurement of Pellets (PBX).

CUSTODIAN:
ARMY-MU

PREPARING ACTIVITY:
ARMY-MU

Project Number: 1376-A-006

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Department of the Army
Headquarters, Picatinny Arsenal
Dover, New Jersey 07801

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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